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### **ELECTRONIC DEVICE AND METHOD FOR CONTROLLING PLATOONING VEHICLES**

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#### **Abstract**

A device for platooning of vehicles includes memory storing instructions and a processor, and the instructions, when executed by the processor, causes the electronic device to identify branch points for each of the vehicles on a route of the platooning, identify distances from current positions of the vehicles to the branch points, determine an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

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## **Background/Summary**

### **CROSS-REFERENCE TO RELATED APPLICATION(S)**

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0031638, filed on Mar. 5, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

### **BACKGROUND**

#### **Technical Field**

[0002] The present disclosure relates to an electronic device and a method for controlling platooning vehicles.

#### **Description of Related Art**

[0003] Platooning is a technology that controls autonomous driving of two or more vehicles. Vehicles forming a platoon may drive while forming a certain formation. The platooning may, by reducing the spacing between the vehicles, improve fuel efficiency by reducing air resistance, reduce a risk of accidents, and reduce traffic congestion by controlling a vehicle flow. The vehicles forming the platoon may include a lead vehicle and following vehicles.

[0004] The above-described information may be provided as a related art for the purpose of helping to understand the present disclosure. No claim or determination is raised as to whether any of the above-described information may be applied as a prior art related to the present disclosure.

### **SUMMARY**

[0005] In an embodiment, an electronic device for platooning of vehicles may comprise memory storing instructions and a processor, and the instructions, when executed by the processor, may cause the electronic device to identify branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon, identify distances from current positions of the vehicles to the branch points, determine an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

[0006] In an embodiment, a method performed by an electronic device for platooning of vehicles may comprise identifying branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon, identifying distances from current positions of the vehicles to the branch points, determining an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, controlling the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

[0007] In an embodiment, a computer-readable medium may store one or more programs. The one or more programs, when executed by an electronic device for platooning of vehicles, may cause the electronic device to identify branch points for each of the vehicles on a route of the platooning,

wherein the branch points are locations where the vehicles leave the platoon, identify distances from current positions of the vehicles to the branch points, determine an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1A schematically illustrates platooning vehicles.

[0009] FIGS. 1B and 1C illustrate an example of a conventional truck.

[0010] FIG. 2 is a block diagram of electronic devices for platooning of vehicles according to an embodiment.

[0011] FIGS. 3A, 3B, and 3C illustrate a formation of vehicles performing platooning according to an embodiment.

[0012] FIGS. 4A, 4B, and 4C illustrate a formation of vehicles performing platooning according to an embodiment.

[0013] FIGS. 5A, 5B, and 5C illustrate a formation of vehicles performing platooning according to an embodiment.

[0014] FIGS. 6A, 6B, and 6C illustrate a formation of vehicles performing platooning according to an embodiment.

[0015] FIGS. 7A and 7B illustrate a formation of vehicles performing platooning according to an embodiment.

[0016] FIG. 8 is a flowchart of an electronic device according to an embodiment.

[0017] FIG. 9 is a flowchart of an electronic device according to an embodiment.

[0018] FIG. 10 is a flowchart of an electronic device according to an embodiment.

[0019] FIG. 11 illustrates an example of a block diagram illustrating an autonomous driving system of a vehicle according to an embodiment.

[0020] FIGS. 12 and 13 illustrate an example of a block diagram indicating an autonomous driving moving object according to an embodiment.

[0021] FIG. 14 illustrates an example of a gateway related to a user device according to various embodiments.

[0022] FIG. 15 is a diagram for explaining an operation of an electronic device for training a neural network based on a set of learning data, according to an embodiment.

[0023] FIG. 16 is a block diagram of an electronic device according to an embodiment.

[0024] FIG. 17 is a flowchart of an electronic device according to an embodiment.

### DETAILED DESCRIPTION

[0025] Specific structural or functional descriptions of embodiments according to a concept of the present invention disclosed in the present specification are illustrated only for a purpose of describing the embodiments according to the concept of the present invention, and the embodiments according to the concept of the present invention may be implemented in various forms and are not limited to embodiments described in the present specification.

[0026] Since the embodiments according to the concept of the present invention may apply various changes and have various forms, the embodiments will be illustrated in the drawings and described in detail in the present specification. However, this is not intended to limit the embodiments according to the concept of the present invention to specific disclosure forms, and includes modifications, equivalents, or substitutes included in a spirit and a technical scope of the present

invention.

[0027] Although terms such as ‘first’ or ‘second’ may be used to describe various components, the components should not be limited by the terms. The terms are only for a purpose of distinguishing one component from another component, for example, without departing from a scope of rights according to the concept of the present invention, a first component may be referred to as a second component, and similarly the second component may also be referred to as the first component.

[0028] When a component is said to be “connected” or “accessed” to another component, it should be understood that it may be directly connected or accessed to the other component, but still another component may exist in a middle. On the other hand, when a component is said to be “directly connected” or “directly connected” to another component, it should be understood that no other component exists in the middle. Expressions that describe a relationship between components, such as “between” and “directly between” or “directly adjacent to”, should be interpreted in the same way.

[0029] A term used in the present specification is used only to describe specific embodiments and is not intended to limit the present invention. Singular expressions include plural expressions unless context clearly indicates otherwise. In the present specification, a term such as “include” or “have”, and the like is intended to be designated as existence of a described feature, number, step, operation, component, part, or a combination thereof, and should be understood not to preclude a probability of the existence or addition of one or more other features, numbers, steps, operations, components, parts, or a combination thereof.

[0030] Unless otherwise defined, all terms used herein, including technical or scientific terms, have the same meaning as commonly understood by a person of ordinary skill in the art to which the present invention belongs. Terms defined in commonly used dictionaries should be interpreted as having a meaning consistent with the meaning of the context of relevant technology, and are not interpreted in an ideal or overly formal sense unless explicitly defined herein.

[0031] Hereinafter, embodiments will be described in detail with reference to the accompanying drawings. However, a scope of a patent application is not limited or restricted by these embodiments. The same reference numerals presented in each drawing may indicate the same configuration, and overlapping descriptions thereof may be omitted.

[0032] FIG. 1A schematically illustrates platooning vehicles.

[0033] Platooning is a technology that controls two or more vehicles **10** forming a platoon to drive while maintaining a designated formation. Each of the vehicles **10** may include electronic devices (e.g., an electronic device **100** and other electronic devices **200** of FIG. 2) for the platooning. The electronic devices **100** and **200** may share control information of the vehicles **10** and information collected through the electronic devices **100** and **200** disposed in each of the vehicles **10** in real time using wireless communication technology. As the wireless communication technology for exchanging information between the electronic devices **100** and **200** illustrated in FIG. 1A, various wireless access technologies such as Vehicle to Everything (V2X) including Vehicle to Infrastructure (V2I), Vehicle to Device (V2D), Vehicle to Vehicle (V2V), Vehicle to Pedestrian (V2P), and the like, 5G New Radio (NR) Sidelink in a cellular manner, and Dedicated Short Range Communication (DSRC) based on 802.11 may be used.

[0034] The vehicles **10** may be distinguished into a vehicle A, which is a lead vehicle, and following vehicles **17**. From among the platooning vehicles **10**, the vehicle A may be referred to as a vehicle located at the front on a traveling route, and the following vehicles **17** may be referred to as remaining vehicles other than vehicle A. The electronic device **100** disposed in the vehicle A may be used to control an overall operation of the platooning. For example, since the vehicle A is disposed at the front in the platoon, the electronic device **100** may obtain more various information than the other electronic devices **200**.

[0035] According to an embodiment, each of the vehicles **10** may be configured based on various shapes. For example, a shape of each of the vehicles **10** may be configured according to a vehicle

type. The vehicle type may include a passenger car, a sports car, a military vehicle, a truck, a bus, a motorcycle, and a forklift. However, it is not limited thereto. For example, the vehicle A may be configured in a passenger car shape (e.g., a sedan). A vehicle B among the following vehicles **17** may be configured in a truck shape including a tractor and a trailer. A vehicle C, a vehicle D, a vehicle E, and a vehicle F among the following vehicles **17** may be configured in the passenger car shape. However, the vehicle type of each of the vehicles **10** is not limited by the above-described example.

[0036] The electronic device **100** may transmit and/or receive data with an external electronic device (e.g., a base station **33**, a satellite **34**, and/or a server **35**). For example, the electronic device **100** may transmit and/or receive data with the server **35** through at least one of the base station **33** and/or the satellite **34**.

[0037] For example, in order to determine a traveling route, the electronic device **100** may receive data including information related to the traveling route from the external electronic device (e.g., the base station **33**, the satellite **34**, and/or the server **35**), and transmit data including information related to a real-time location of the platoon to the external electronic device (e.g., the base station **33**, the satellite **34**, and/or the server **35**).

[0038] According to an embodiment, the base station **33** and/or the server **35** may be configured to manage the platooning of a designated area. For example, the base station **33** and/or the server **35** may be configured to manage traveling (or the platooning) of vehicles located within a cell defined based on coverage. The vehicles may be controlled through another base station and/or another server whenever the cell is changed. The vehicles may establish a connection (e.g., handover) with the other base station whenever the cell is changed.

[0039] According to an embodiment, the electronic device **100** may be configured to control the traveling of the vehicles **10** based on information (e.g., a traveling route, a traveling speed, a space between the vehicles **10**, and/or a formation of the platoon) related to the platooning vehicles **10** and/or information (e.g., a road condition, other vehicles **20**, a lane **30**, and/or lanes **40** including a lane **41** and a lane **42**) related to a surrounding environment. For example, the electronic device **100** may transmit a signal for controlling the platooning to each of the other electronic devices **200** disposed in each of the following vehicles **17**. The other electronic devices **200** may be configured to control the traveling of the following vehicles **17** based on the signal received from the electronic device **100**.

[0040] Each of the vehicles **10** may include various types of vehicles. The vehicles **10** may include a platooning section. Each of the vehicles **10** may pass a branch point leaving from the platooning section, according to a destination. Due to the leaving of any one vehicle in the middle of the platoon formed by the vehicles **10**, a speed of the platooning of the vehicles **10** may be slowed or a probability of an accident may increase. In the following specification, a technical feature for controlling the vehicles **10** performing the platooning will be described.

[0041] Hereinafter, the electronic device **100** for controlling the vehicles **10** performing the platooning will be described with reference to the drawings. In the present disclosure, terms such as a first lane, a second lane, are simply used to distinguish lanes and are not limited to any further meaning.

[0042] FIGS. **1B** and **1C** illustrate an example of a conventional truck. For many years, a trucking industry has experienced steady growth and has expanded a range of services to respond to more complex supply chains. These services include last-mile deliveries, drop-trailer programs, and intermodal transportation (a transportation type in which freight is carried to a destination by two or more different means of transportation (a ship and a rail, a ship and an airplane)) at a port.

[0043] As such, since there are so many different ways to carry the freight, manufacturers of freight-related equipment have designed different types of equipment to carry the freight according to various transportation needs.

[0044] In the present specification, a truck that tows a trailer for a main purpose of a freight carry

(or catering) is collectively referred to as a tractor.

[0045] The tractor described in the present specification may be classified into a conventional truck (or a bonneted truck), a cab-over truck (or a cab-over engine), and a semi-conventional truck, which is an intermediate form between the conventional truck and the cab-over truck, according to a location and a form of a cab of the tractor.

[0046] The conventional truck, which is a form in which an engine and a hood are positioned above a front axle in front of a cab of a tractor, and has a structure in which a driver sits behind the front axle, is a type of a tractor mainly used in North America where the engine of the tractor is located in front of the driver.

[0047] On the other hand, the cap over truck, which has a structure in which a cap of a tractor is located to a front end of the tractor and a driver sits in front of a front axle, and is a form a so-called “flat face (or flat nose)” in which a front of the tractor is flat, is a form of a tractor mainly used in most countries, such as Europe and Asia where an engine of the tractor is located under the driver.

[0048] Just as there are various forms according to a purpose and a demand of a tractor, there are various forms of trailers towed by the tractor. Among them, the most representative types of trailers are a full-trailer and a semi-trailer. The full trailer and the semi-trailer may be distinguished by whether the trailer is equipped with both a front axle and a rear axle. This trailer may be connected to a box truck or a tractor through a coupling device.

[0049] To explain specifically, the full trailer is a commercial freight trailer equipped with both the front axle and the rear axle. The full trailer may fully support its own weight without relying on a tractor by being to support an entire load only with the trailer, is equipped with a drawbar to be coupled to a towing unit (or a hauling unit) such as the tractor, and is mainly used in the United States and Canada and the like.

[0050] On the other hand, the semi-trailer, which is a freight trailer with only a rear axle without a front axle, may support a large portion of a load by a tractor connected by a kind of hitch called a “fifth wheel” (custom-character). In case of being in a stationary state by being detached from the tractor, the semi-trailer may support the load of the trailer by spreading a landing gear mounted on a lower portion of the semi-trailer vertically to the ground. A combination of the semi-trailer and the tractor is called a semi-trailer truck (also called simply a “semi-trailer”, a “tractor-trailer”, a “semi-truck”, a “big rig”, or a “semi” in the United States). The above-described “Fifth wheel” refers to a horizontal wheel attached to a tractor axle of a trailer truck to facilitate a direction change of the trailer, and is also called a fifth wheel. The “Fifth wheel” is a device that enables a movable coupling (custom-character) of the tractor and the semi-trailer and includes a lower portion consisting of a trunnion plate and a clasp (custom-character) device that firmly fixes a kingpin mounted on the semi-trailer to the tractor.

[0051] Hereinafter, in the present specification, based on the terms of the tractors/trailers described above, for convenience of explanation, “trailer” will be used to mean a freight transport vehicle connected to a tractor for a trailer, and “trailer” will be used as a tow vehicle for moving the trailer. In addition, in the present invention, in order to exclude a limitation of rights according to an embodiment described in a detailed description as much as possible, a tractor hauling/towing a “trailer” may be described interchangeably with a term “towing vehicle,” a trailer towed by a tractor may be described interchangeably with a term “towed vehicle”.

[0052] Also, for convenience of explanation, it is desirable to understand that the “trailer” described throughout the present specification refers to a “semi-trailer,” but is not limited thereto.

[0053] Referring to FIGS. 1B and 1C, a vehicle **1015** may be an example of the vehicle B of FIG. 1A described above. The vehicle **1015** may include a tractor or tractor unit **1051** and a semi-trailer **1052**. FIG. 1B illustrates a state in which the tractor **1051** and the semi-trailer **1052** are not connected, and FIG. 1C illustrates a state in which the tractor **1051** and the semi-trailer **1052** are connected.

[0054] In an embodiment, the semi-trailer **1052** may be selectively connected by a fifth wheel hitch

**1056** carried by the tractor **1051**, and the fifth wheel hitch **1056** may be fastened to the kingpin **1058** fixed to the semi-trailer **1052** according to a known manner. The vehicle **1015** including the tractor **1051** and the semi-trailer **1052** may be referred to as a truck. The vehicle **1015** may include only the tractor **1051**. The semi-trailer **1052** illustrated in FIGS. **1B** and **1C** illustrated a form of the “semi-trailer”, but this is for convenience of explanation, and an embodiment of the present disclosure should not be understood as being applied only to the form of the “semi-trailer”. The tractor **1051** illustrated in FIGS. **1B** and **1C** illustrated a form of the “cap-over truck,” but this is for convenience of explanation, and the embodiment of the present disclosure should not be understood as being applied only to the form of the “cap-over truck.”

[0055] In an embodiment, the semi-trailer **1052** may include the king pin **1058** coupled to the fifth wheel hitch **1056** of the tractor **1051**, and a landing gear **1059** supporting the semi-trailer **1052** from the ground in a state in which the semi-trailer **1052** is not coupled to the tractor **1051**. The king pin **1058** and the landing gear **1059** may be installed (or disposed) in a lower portion of the semi-trailer **1052**.

[0056] In an embodiment, the semi-trailer **1052** may be rotatably coupled to the tractor **1051** to support traveling on a curved road. For example, the tractor **1051** and the semi-trailer **1052** may be rotatably coupled through a coupling device including the fifth wheel hitch **1056** and the king pin **1058**. However, a link mechanism between the tractor **1051** and the semi-trailer **1052** is not limited thereto.

[0057] FIG. **2** is a block diagram of electronic devices for platooning of vehicles according to an embodiment.

[0058] Referring to FIG. **2**, an electronic device **100** according to an embodiment may include a processor **110**, memory **120**, a wireless communication device **130**, a camera **140**, a Global Positioning System (GPS) sensor **150**, and/or a wired communication device **160**. The electronic device **100** according to an embodiment may be referred to as an electronic device disposed in a lead vehicle (e.g., the vehicle A of FIG. **1A**).

[0059] For example, the processor **110**, the memory **120**, the wireless communication device **130**, the camera **140**, the GPS sensor **150**, and/or the wired communication device **160** may be electronically and/or operably coupled with each other by an electrical component such as a communication bus. Hereinafter, hardware being operatively coupled may mean that a direct connection or an indirect connection between the hardware is established by wire or wirelessly so that second hardware is controlled by first hardware among the hardware.

[0060] In FIG. **2**, the processor **110**, the memory **120**, the camera **140**, the wireless communication device **130**, the GPS sensor **150**, and/or the wired communication device **160** are illustrated in different blocks, but are not limited thereto. Some of the hardware illustrated in FIG. **2** may be implemented as a single integrated circuit or as portion of a single package, such as a system on a chip (SoC).

[0061] According to an embodiment, the memory **120** may store instructions. The processor **110** may be configured to process data based on the instructions stored in the memory **120**. For example, the processor **110** may include an arithmetic and logic unit (ALU), a floating point unit (FPU), a field programmable gate array (FPGA), a central processing unit (CPU), and/or an application processor (AP). The processor **110** may have a structure of a single-core processor **110**, or may have a structure of a multi-core processor such as a dual core, a quad core, a hexa core, or an octa core.

[0062] According to an embodiment, the memory **120** may include a hardware component for storing data and/or instructions that may be executed by the processor **110**. For example, the memory **120** may include a volatile memory such as random-access memory (RAM) and/or a non-volatile memory such as read-only memory (ROM). For example, the volatile memory may include at least one of dynamic RAM (DRAM), static RAM (SRAM), cache RAM, and pseudo SRAM (PSRAM). For example, the non-volatile memory may include at least one of programmable ROM

(PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), flash memory, a hard disk, a compact disk, a solid state drive (SSD), and an embedded multi-media card (eMMC). For example, the memory **120** of the electronic device **100** may include a neural network model. The electronic device **100** may identify an external object (e.g., a lane (e.g., the lane **30** of FIG. **1A**), a lane (e.g., the lanes **40** of FIG. **1A**), other vehicles (e.g., the other vehicles **20** of FIG. **1A**) and/or traffic lights (e.g., the traffic light **50** of FIG. **1A**) based on the neural network model stored in the memory **120**.

[0063] According to an embodiment, the wireless communication device **130** may be used for wireless communication with other electronic devices **200** and/or an external electronic device. For example, the electronic device **100** may be configured to wirelessly communicate with an external electronic device (e.g., a base station (e.g., the base station **33** of FIG. **1A**), a satellite (e.g., the satellite **34** of FIG. **1A**), and/or a server (e.g., the server **35** of FIG. **1A**)) and the other electronic devices **200** using the wireless communication device **130**. The wireless communication device **130** may be electrically connected with an antenna (e.g., an antennas **1332a** or **1332b** of FIG. **13**) for transmitting and/or receiving a signal. The wireless communication device **130** may convert an analog signal provided from the processor **110** into a digital signal and upconvert a baseband signal into a radio frequency (RF) signal. The electronic device **100** may obtain information related to a real-time location of a platoon using the GPS sensor **150**, and transmit data including the information to the external electronic device using the wireless communication device **130**. The electronic device **100** may transmit a signal for controlling traveling of following vehicles (e.g., the following vehicles **17** of FIG. **1A**) to a wireless communication device **230** of the other electronic devices **200**. The other electronic devices **200** may receive the signal through the wireless communication device **230**.

[0064] According to an embodiment, the camera **140** may include a lens assembly or an image sensor. The lens assembly may collect light emitted from a subject which is a target of image capturing. The lens assembly may include one or more lenses. For example, the camera **140** may include a plurality of lens assemblies. For example, in the camera **140**, a portion the plurality of lens assemblies may have identical lens properties (e.g., an angle of view, a focal length, an autofocus, an f number, or an optical zoom), or at least one lens assembly may have one or more lens properties different from lens properties of another lens assembly. The lens assembly may include a wide-angle lens or a telephoto lens. For example, the electronic device **100** may include a flash for the camera **140**. The flash may include one or more light emitting diodes (e.g., a red-green-blue (RGB) LED, a white LED, an infrared LED, or an ultraviolet LED), or a xenon lamp. For example, the image sensor may obtain an image corresponding to the subject by converting light emitted or reflected from the subject and transmitted through the lens assembly into an electrical signal. According to an embodiment, the image sensor may include, for example, one image sensor selected from among image sensors in which a property is different, such as an RGB sensor, a black and white (BW) sensor, an IR sensor, or a UV sensor, a plurality of image sensors having an identical property, or a plurality of image sensors having a different property. Each image sensor included in the image sensor may be implemented, for example, using a charged coupled device (CCD) sensor or a complementary metal oxide semiconductor (CMOS) sensor.

[0065] According to an embodiment, the electronic device **100** may identify surrounding environments of the vehicle A using the camera **140**. For example, the electronic device **100** may identify an external object based on an image obtained through camera **140**. For example, the electronic device **100** may identify the external object corresponding to the image from the image obtained through the camera **140** using the neural network model. For example, the electronic device **100** may obtain an image corresponding to the other vehicles **20** traveling in another lane (e.g., the lane **42** of FIG. **1A**) through the camera **140**, and identify the other vehicles **20** in the lane **42** from the image.

[0066] According to an embodiment, the wired communication device **160** may be used to connect



the electronic device **100** and control circuitry (e.g., an electronic control unit (ECU)) of the vehicle A. For example, the electronic device **100** may transmit a signal for controlling the vehicle A to the control circuitry of the vehicle A using the wired communication device **160**. The electronic device **100** may control the vehicle A using the control circuitry connected with the wired communication device **160**.

[0067] The other electronic devices **200** disposed in the following vehicles **17** may include substantially the same components as the electronic device **100** disposed in the vehicle A. For example, each of the other electronic devices **200** may include a processor **210**, memory **220**, a wireless communication device **230**, a camera **240**, a GPS sensor **250**, and/or a wired communication device **260**. The above-described descriptions with respect to components of the electronic device **100** may be applied to components of the other electronic devices **200** in substantially the same manner.

[0068] Since the vehicles **10** drive in a designated formation, the camera **240** of the other electronic devices **200** may obtain an image that the camera **140** of the electronic device **100** cannot obtain, at a specific timing. According to an embodiment, the other electronic devices **200** may transmit information related to the image obtained through the camera **240** and/or information related to the external object identified from the image, to the electronic device **100**. The electronic device **100** may identify the surrounding environments of the platoon based on the information received from the other electronic devices **200**, and control the traveling of the platoon based on the surrounding environments.

[0069] According to an embodiment, the electronic device **100** may change the formation using the neural network model. For example, the processor **110** may determine whether to change the formation based on information (e.g., first environment information) related to the surrounding environments of the vehicle A and information (e.g., second environment information) received from the following vehicles **17**, obtained using the camera **140**. For example, on a travel route for the platooning, in case that a traffic flow is not smooth, the processor **110** may change the formation or a traveling method (e.g., a speed, the travel route) for the platooning. For example, the processor **110** may change the formation or the traveling method for the platooning, based on a state (e.g., an amount of remaining fuel, a tire pressure, or an engine oil pressure) of the vehicles **10**.

[0070] FIGS. 3A, 3B, and 3C illustrate a formation of vehicles performing platooning according to an embodiment.

[0071] FIGS. 3A, 3B, and 3C illustrate vehicles **10** as time elapses. For example, FIG. 3A illustrates the vehicles **10** at a first timing, FIG. 3B illustrates the vehicles **10** at a second timing, and FIG. 3C illustrates the vehicles **10** at a third timing after the second timing.

[0072] Referring to FIGS. 3A, 3B, and 3C, in order to perform the platooning, the vehicles **10** may configure a platoon **310**. For example, the platoon **310** may include vehicles A, B, C, D, E, and F. Although an example including six vehicles is illustrated, the number of the vehicles **10** is not limited thereto, and the number of the vehicles **10** may be variously set. A shape of the vehicles **10** is for convenience of explanation and is not limited by the illustrated example.

[0073] Operations described below may be performed by an electronic device (e.g., the electronic devices **100** and **200**) and/or a processor (e.g., the processors **110** and **210**) included in each of the vehicles **10**. However, for convenience, it has been described that the following operations are performed by each of the vehicles **10**.

[0074] In an embodiment, the vehicle A may control the vehicles **10** based on transmitting a signal to other vehicles B, C, D, E, and F. In an example, the vehicle A may control the vehicle B based on transmitting a signal to the other vehicles B, C, D, E, and F. For example, the vehicle A may control the vehicle C based on transmitting a signal to the other vehicles B, C, D, E, and F. For example, the vehicle A may control the vehicle D based on transmitting a signal to the other vehicles B, C, D, E, and F. For example, the vehicle A may control the vehicle E based on

transmitting a signal to the other vehicles B, C, D, E, and F. For example, the vehicle A may control the vehicle F based on transmitting a signal to the other vehicles B, C, D, E, and F.

[0075] In the following specification, for convenience of explanation, an operation in which the vehicle A controls the vehicles **10** by transmitting a signal to the other vehicles B, C, D, E, and F will be described as an operation in which the vehicle A controls the vehicles **10**.

[0076] According to an embodiment, the vehicle A may control the vehicles **10** to perform the platooning. The vehicle A may configure the vehicles **10** as a group. The vehicle A may control the configured group to move to an identical route. For example, the vehicle A may identify a route in which the vehicles **10** will travel to a destination.

[0077] In an embodiment, the vehicle A may identify branch points of each of the vehicles **10** on a route of the platooning. For example, the vehicle A may obtain information regarding destinations of the vehicles **10** from each of the vehicles **10** and, based on the obtained information, identify a branch point of each of the vehicles **10**. The branch point may be a location where the vehicle leaves the platoon **310**. For example, the branch point may be a point (or location) where a vehicle leaves from a route of the platooning to another route.

[0078] Referring to FIG. 3A, in an embodiment, the vehicle A may identify distances from current locations of the vehicles **10** to the branch points. For example, a distance to a branch point of the vehicle A may be 10 km. For example, a distance to a branch point of the vehicle B may be 5 km. For example, a distance to a branch point of the vehicle C may be 30 km. For example, a distance to a branch point of the vehicle D may be 3 km. For example, a distance to a branch point of the vehicle E may be 50 km. For example, a distance to a branch point of the vehicle F may be 60 km. However, a distance to the branch points of the illustrated vehicles **10** is only for explaining a scenario for controlling the platooning, and the branch points of the vehicles **10** are not limited by the example.

[0079] The route illustrated in FIG. 3A may be composed of two lanes **40**. The two lanes **40** may include a lane **41** and a lane **42**. At the timing of FIG. 3A, the vehicles **10** may be in a state of traveling in the lane **41**.

[0080] In an embodiment, the vehicle A may rearrange an order of the vehicles **10** configuring the platoon **310**. For example, the vehicle A may determine the order of the vehicles **10** in platoon **310**, based on an ascending order of distances to the branch point.

[0081] In an embodiment, the vehicle A may identify at least one vehicle that has to leave a lane in which the platoon **310** is traveling in order for the vehicles **10** to be rearranged according to the determined order, based on an insertion sort algorithm. For example, in the example of FIG. 3A, since the vehicle B has a smaller distance to the branch point than the vehicle A, which is a preceding vehicle, and the vehicle D has a smaller distance to the branch point than the vehicle C, which is a preceding vehicle, the vehicles B and D may have to leave the lane **41** in which the platoon **310** is traveling for rearrangement.

[0082] In an embodiment, the vehicle A may identify whether there are vehicles whose distances to the branch points are less than a reference value among vehicles in which leaving for rearrangement is required. For a non-limiting example, the reference value may be 1 km. In an embodiment, the vehicle A may change the determined order such that a corresponding vehicle(s) is arranged in the ascending order of distances from a tail to the branch point, based on identifying the vehicle(s) among the vehicles **10** whose distances to the branch points are less than the reference value. In the example of FIG. 3A, among the vehicles B and D that need to leave the platoon **310** for the rearrangement, a vehicle less than the reference value may not exist. Accordingly, the vehicle A may determine the order of the vehicles **10** in an order continuing from a head to the vehicles D, B, A, C, E, and F.

[0083] In an embodiment, the vehicle A may determine whether a lead vehicle (e.g., the vehicle A) in a current order of the vehicles **10** in the platoon **310** is different from a lead vehicle (e.g., the vehicle D) in the determined (or changed) order and/or whether a tail vehicle (e.g., the vehicle F) in

the current order of the vehicles **10** in the platoon **310** is different from a tail vehicle (e.g., the vehicle F) in the determined (or changed) order.

[0084] In an embodiment, in case of determining that the lead vehicle is different or/or the tail vehicle is different, the vehicle A may control the vehicles **10** to rearrange the vehicles **10** in the platoon **310** in the determined order. Alternatively, in case that both the lead vehicle and the tail vehicle are not changed, the vehicle A may control the vehicles **10** so that the current order of the vehicles **10** in the platoon **310** is maintained.

[0085] In the example of FIG. 3A, a tail vehicle of the platoon **310** is the same as the vehicle F, but since a lead vehicle changes from the vehicle A to the vehicle D, the vehicles **10** may control the vehicles **10** to be rearranged in an order of the vehicles D, B, A, C, E, and F from the lead vehicle.

[0086] In an embodiment, the vehicle A may determine the at least one vehicle that needs to leave the platoon **310** as a first group where a distance to the branch point exceeds the reference value (e.g., 1 km) and/or as a second group where a distance to the branch point does not exceed the reference value. In the example of FIG. 3A, since the distances to the branch points of the vehicles B and D that need to leave the platoon **310** are 5 km and 3 km, which exceed the reference value, the vehicles B and D may be determined as the first group.

[0087] In an embodiment, in case that the first group includes a first plurality of vehicles, the vehicle A may control the vehicles **10** such that the first plurality of vehicles form a first sub-platoon in another lane different from a lane in which the platoon **310** is traveling, and subsequently merge at a head of the platoon **310**.

[0088] In an embodiment, in case that the first group includes only one vehicle, the vehicle A may control the vehicles **10** such that a corresponding vehicle leaves the lane in which the platoon **310** is traveling and subsequently merges at a head of the platoon.

[0089] In the example of FIG. 3A, the first group may include the plurality of vehicles B and D. Therefore, referring to FIGS. 3B and 3C, the vehicle A may control the vehicles **10** such that the vehicles B and D form a first sub-platoon **311** in another lane **42** different from the lane **41** in which the platoon **310** is traveling, and subsequently merge at the head of the platoon **310**.

[0090] FIGS. 4A, 4B, and 4C illustrate a formation of vehicles performing platooning according to an embodiment.

[0091] FIGS. 4A, 4B, and 4C illustrate vehicles **10** as time elapses. For example, FIG. 4A illustrates the vehicles **10** at a first timing, FIG. 4B illustrates the vehicles **10** at a second timing, and FIG. 4C illustrates the vehicles **10** at a third timing after the second timing.

[0092] Referring to FIG. 4A, a distance to a branch point of a vehicle A may be 10 km. For example, a distance to a branch point of a vehicle B may be 800 m. For example, a distance to a branch point of a vehicle C may be 30 km. For example, a distance to a branch point of a vehicle D may be 900 m. For example, a distance to a branch point of a vehicle E may be 50 km. For example, a distance to a branch point of a vehicle F may be 60 km. However, a distance to the branch points of the illustrated vehicles **10** is only for explaining a scenario for controlling the platooning, and the branch points of the vehicles **10** are not limited by the example.

[0093] In an embodiment, the vehicle A may determine an order of the vehicles **10** as the vehicles B, D, A, C, E, and F according to an ascending order of the distances to the branch point.

[0094] In an embodiment, vehicles that have to leave for rearrangement among the vehicles **10** may be the vehicles B and D. The vehicles D and B have a distance to the branch point less than or equal to a reference value (e.g., 1 km). Accordingly, the vehicle A may change the order of the vehicles **10** such that the vehicles B and D may be arranged in the ascending order of distances from a tail of a platoon **310** to the branch point. The changed order may be the vehicles A, C, E, F, D, and B from a head.

[0095] In an embodiment, by the change of the order of the vehicles **10**, a lead vehicle may be identical as the vehicle A, but a tail vehicle may be changed from the vehicle F to the vehicle B. Accordingly, the vehicle A may control the vehicles **10** to be rearranged in the order of the vehicles

A, C, E, F, D, and B from the head.

[0096] In an embodiment, the vehicle A may determine the vehicles B and D that need to leave the platoon **310** as a second group where a distance to the branch point does not exceed the reference value.

[0097] In an embodiment, in case that the second group includes a second plurality of vehicles, the vehicle A may control the vehicles **10**, such that the second plurality of vehicles form a second sub-platoon in another lane different from a lane in which the platoon **310** is traveling, and subsequently merge at the tail of the platoon **310**.

[0098] In an embodiment, in case that the second group includes only one vehicle, the vehicle A may control the vehicles such that a corresponding vehicle merges at the head of the platoon **310** in the lane in which the platoon **310** is traveling.

[0099] In the example of FIG. **4A**, the second group may include the plurality of vehicles B and D. Therefore, referring to FIGS. **4B** and **4C**, the vehicle A may control the vehicles **10** such that the vehicles B and D form a second sub-platoon **411** in another line **42** different from a lane **41** in which the platoon is traveling, and subsequently merge at the tail of the platoon **310**.

[0100] FIGS. **5A**, **5B**, and **5C** illustrate a formation of vehicles performing platooning according to an embodiment.

[0101] FIGS. **5A**, **5B**, and **5C** illustrate vehicles **10** as time elapses. For example, FIG. **5A** illustrates the vehicles **10** at a first timing, FIG. **5B** illustrates the vehicles **10** at a second timing after the first timing, and FIG. **5C** illustrates the vehicles **10** at a third timing after the timing.

[0102] Referring to FIG. **5A**, a distance to a branch point of a vehicle A may be 10 km. For example, a distance to a branch point of a vehicle B may be 5 km. For example, a distance to a branch point of a vehicle C may be 30 km. For example, a distance to a branch point of a vehicle D may be 800 m. For example, a distance to a branch point of a vehicle E may be 900 m. For example, a distance to a branch point of a vehicle F may be 60 km. However, a distance to the branch points of the illustrated vehicles **10** is only for explaining a scenario for controlling the platooning, and the branch points of the vehicles **10** are not limited by the example.

[0103] In an embodiment, the vehicle A may determine an order of the vehicles **10** as the vehicles D, E, B, A, C, and F according to an ascending order of the distances to the branch point.

[0104] In an embodiment, the vehicles have to leave for rearrangement among the vehicles **10** may be the vehicles B, D, and E. The vehicles D and E may have a distance to the branch point less than or equal to a reference value (e.g., 1 km), and the vehicle B may have a distance to the branch point exceeding the reference value. Accordingly, the vehicle A may change the order of the vehicles **10** such that the vehicles D and E may be arranged in the ascending order of distances from a tail of a platoon **310** to the branch point. The changed order may be the vehicles B, A, C, F, D, and E from a head.

[0105] A lead vehicle was changed from the vehicle A to the vehicle B, and the tail vehicle may be changed from the vehicle F to the vehicle E, by the change of the order of the vehicles **10**.

Accordingly, the vehicle A may control the vehicles **10** to be rearranged in the order of the vehicles B, A, C, F, D, and E from the head.

[0106] In an embodiment, the vehicles **10** may determine a vehicle that needs to leave the platoon **310** as a first group where a distance to the branch point exceeds the reference value and/or as a second group where a distance to the branch point does not exceed the reference value.

[0107] In an embodiment, the vehicle A may determine the vehicles D and E as the second group where the distance to the branch point is less than or equal to the reference value. In addition, vehicle A may determine the vehicle B as the first group where the distance to the branch point exceeds the reference value.

[0108] Referring to FIGS. **5B** and **5C**, in an embodiment, since the first group includes only the one vehicle B, the vehicle A may control the vehicles **10** such that a corresponding vehicle leaves a lane **41** in which the platoon **310** is traveling and subsequently merges at the head of the platoon.

[0109] In addition, the vehicle A may control vehicles **10** such that the vehicles D and E form a third sub-platoon **511** in another lane **42** different from the lane **41** in which the platoon **310** is traveling, and subsequently merge at the tail of the platoon **310**.

[0110] FIGS. **6A**, **6B**, and **6C** illustrate a formation of vehicles performing platooning according to an embodiment.

[0111] FIGS. **6A**, **6B**, and **6C** illustrate vehicles **10** as time elapses. For example, FIG. **6A** illustrates the vehicles **10** at a first timing, FIG. **6B** illustrates the vehicles **10** at a second timing, and FIG. **6C** illustrates the vehicles **10** at a third timing after the second timing.

[0112] Referring to FIG. **6A**, a distance to a branch point of a vehicle A may be 5 km. For example, a distance to a branch point of a vehicle B may be 3 km. For example, a distance to a branch point of a vehicle C may be 3 km. For example, a distance to a branch point of a vehicle D may be 0.3 km. For example, a distance to a branch point of a vehicle E may be 0.1 km. For example, a distance to a branch point of a vehicle F may be 0.4 km. However, a distance to the branch points of the illustrated vehicles **10** is only for explaining a scenario for controlling the platooning, and the branch points of the vehicles **10** are not limited by the example.

[0113] In an embodiment, the vehicle A may determine an order of the vehicles **10** as the vehicles E, D, F, B, C, and A according to an ascending order of the distances to the branch point.

[0114] In an embodiment, vehicles that have to leave for rearrangement among the vehicles **10** may be the vehicles A, B, D, E, and F. The vehicles D, E, and F may have a distance to the branch point less than or equal to a reference value (e.g., 1 km), and the vehicles A and B may have a distance to the branch point exceeding the reference value. Accordingly, the vehicle A may change the order of the vehicles **10** such that the vehicles D, E, and F may be arranged in the ascending order of distances from a tail of a platoon **310** to the branch point. The changed order may be the vehicles B, C, A, F, D, and E from a head.

[0115] A lead vehicle was changed from the vehicle A to the vehicle B, and a tail vehicle may be changed from the vehicle F to the vehicle E, by the change of the order of the vehicles **10**. Accordingly, the vehicle A may control the vehicles **10** to be rearranged in the order of the vehicles B, C, A, F, D, and E from the head.

[0116] In an embodiment, the vehicle A may determine the vehicles A and B where a distance to the branch point exceeds the reference value (e.g., 1 km) as a first group, and determine the vehicles D, E, and F where a distance to the branch point does not exceed the reference value (e.g., 1 km) as a second group.

[0117] In an embodiment, in case that the first group or the second group includes a third plurality of vehicles exceeding a reference number, the vehicle A may identify a specific vehicle having the largest distance to the branch point among corresponding vehicles. The reference number may be, for a non-limiting example, three.

[0118] In an embodiment, the vehicle A may control the vehicles **10** such that the specific vehicle maintains the platooning, and may control the vehicles **10** such that remaining vehicles may form a sub-platoon in another lane **42** different from a lane **41** in which the platoon **310** is traveling, and subsequently merge at the tail of the platoon **310**.

[0119] In an embodiment, the first group may include the vehicles A and B that do not exceed the reference number. Accordingly, referring to FIGS. **6A** and **6B**, the vehicle A may control the vehicles **10** such that the vehicles A and B leave the lane **41** in which the platoon **310** is traveling and subsequently merge at the platoon **310** in the determined order.

[0120] In an embodiment, the second group may include the vehicles D, E, and F exceeding the reference number. The vehicle A may identify the vehicle F having the farthest distance to the branch point among the vehicles D, E, and F as the specific vehicle. Accordingly, the vehicle A may control the vehicles **10** such that the vehicle F maintains the platooning in the lane **41**, and the vehicles D and E form a fourth sub-platoon **611** in the other lane **42** different from the lane **41** in which the platoon **310** is traveling, and subsequently merge at the tail of the platoon **310**.

[0121] FIGS. 7A and 7B illustrate a formation of vehicles performing platooning according to an embodiment.

[0122] FIGS. 7A and 7B illustrate vehicles **10** as time elapses. For example, FIG. 7A illustrates the vehicles **10** at a first timing, and FIG. 7B illustrates the vehicles **10** at a second timing after the first timing.

[0123] Referring to FIG. 7A, a distance to a branch point of a vehicle A may be 5 km. For example, a distance to a branch point of a vehicle B may be 3 km. For example, a distance to a branch point of a vehicle C may be 3 km. For example, a distance to a branch point of a vehicle D may be 0.3 km. For example, a distance to a branch point of a vehicle E may be 0.1 km. For example, a distance to a branch point of a vehicle F may be 0.4 km. However, a distance to the branch points of the illustrated vehicles **10** is only for explaining a scenario for controlling the platooning, and the branch points of the vehicles **10** are not limited by the example.

[0124] In an embodiment, the vehicle A may control the vehicles **10** in a platoon **310** to travel along a route **710**.

[0125] In an embodiment, the vehicle A may identify the closest vehicle to the branch point from among the vehicles **10**. For example, the vehicle A may identify the closest vehicle E to a branch point **715**.

[0126] In an embodiment, the vehicle A may confirm whether there is a section including multiple lanes in a route to the branch point of the closest vehicle. In case that there are the multiple lanes, rearrangement operations of the vehicles **10** in the above-described platoon **310** may be performed.

[0127] Alternatively, based on confirming that there is no section including the multiple lanes in the route, the vehicle A may determine a safety speed for the closest vehicle to leave the platoon **310** at the branch point. In FIGS. 7A and 7B, since only a lane **43** exists until the branch point **715**, the vehicle A may confirm that there is no section including the multiple lanes in the route to the branch point **715** of the closest vehicle E. Accordingly, the vehicle A may determine the safety speed at which the vehicle E leaves the lane **710** at the branch point **715**. For example, the vehicle A may determine the safety speed based on a sign **730** that provides speed information on a route **720** after the vehicle E leaves the platoon **310**. For example, the vehicle A may receive the information on the safety speed from the vehicle E. For example, the vehicle A may determine the safety speed based on information on a location corresponding to the branch point **715** and/or the route **720** obtained through map data.

[0128] In an embodiment, the vehicle A may control the vehicles **10** in the platoon **310** to travel at the safe speed until the vehicle E leaves the route **710**.

[0129] In an embodiment, the vehicle A may control the vehicles **10** to normally travel in the platoon after the vehicle E moves to the route **720** along a branch lane **44** of the lane **43**.

[0130] FIG. 8 is a flowchart of an electronic device according to an embodiment.

[0131] Operations described with reference to the following drawings may be performed by an electronic device **100** and/or a processor **110**. Each operation of FIG. 8 may be performed sequentially, but is not necessarily performed sequentially. For example, an order of each operation may be changed, and at least two operations may be performed in parallel.

[0132] Referring to FIG. 8, in operation **810**, the electronic device **100** may identify branch points of each of vehicles **10** on a route of platooning.

[0133] In operation **820**, the electronic device **100** may identify distances from current positions of the vehicles **10** in a platoon **310** to the branch points.

[0134] In operation **830**, the electronic device **100** may determine an order of the vehicles **10** in the platoon **310**, based on an ascending order of the identified distances.

[0135] In operation **840**, the electronic device **100** may change the determined order such that other vehicles are arranged in the ascending order of the identified distances from a tail based on identifying the other vehicles whose distances to the branch points among the vehicles **10** in the platoon **310** are less than a reference value. The operation of identifying whether the distances to

the branch points are less than the reference value may be performed for vehicles that have to leave the platoon **310** for rearrangement among the vehicles **10**.

[0136] In operation **850**, the electronic device **100** may determine whether a comparison result satisfies a specific condition by comparing a current order and the determined order of the vehicles **10** in the platoon **310**. For example, in case that a lead vehicle and/or a tail vehicle change before and after the change of the order, the electronic device **100** may determine that the specific condition has been satisfied. In case that there is no change between the lead vehicle and the tail vehicle before and after the change of the order, the electronic device **100** may determine that the specific condition has not been satisfied.

[0137] In operation **850**, in case that it is determined that the comparison result satisfies the specific condition (operation **860**: YES), operation **870** may be performed, otherwise (operation **860**: NO), operation **880** may be performed.

[0138] In the operation **870**, the electronic device **100** may control the vehicles such that the vehicles **10** in the platoon **310** are rearranged in the determined order.

[0139] In operation **880**, the electronic device **100** may control the vehicles **10** such that the current order of the vehicles **10** is maintained.

[0140] FIG. **9** is a flowchart of an electronic device according to an embodiment.

[0141] Each operation of FIG. **9** may be sequentially performed, but is not necessarily performed sequentially. For example, an order of each operation may be changed, and at least two operations may be performed in parallel.

[0142] Referring to FIG. **9**, in operation **910**, an electronic device **100** may identify at least one vehicle that has to leave a platoon **310** to be rearranged in a determined order based on an insertion sort algorithm of identified distances. Additionally, the operation **910** may be performed to determine a target vehicle for determining whether the distances to the branch point in the operation **840** of FIG. **8** are less than a reference value.

[0143] In operation **920**, the electronic device **100** may determine the at least one vehicle (e.g., a vehicle that needs to leave the platoon **310**) as a first group where a distance to a branch point exceeds the reference value (e.g., 1 km) and/or as a second group where a distance to the branch point does not exceed the reference value.

[0144] In operation **930**, the electronic device **100** may control vehicles **10** based on the number of vehicles included in each of the first group and/or the second group.

[0145] For example, in case that the first group includes a first plurality of vehicles, the electronic device **100** may control the vehicles **10** such that the first plurality of vehicles form a sub-platoon on another lane different from a lane in which the platoon **310** is traveling and subsequently merge at a head of the platoon **310**. For example, in case that the first group includes only one vehicle, the electronic device **100** may control the vehicles **10**, such that the one vehicle leaves the lane in which the platoon **310** is traveling, and subsequently merges at the head of the platoon **310**. For example, in case that the second group includes a second plurality of vehicles, the electronic device **100** may control the vehicles **10** such that the second plurality of vehicles form a sub-platoon in the other lane different from the lane in which the platoon **310** is traveling, and subsequently merge at a tail of the platoon **310**. For example, in case that the second group includes only one vehicle, the electronic device **100** may control the vehicles **10** such that the one vehicle leaves the lane in which the platoon **310** is traveling, and subsequently merges at the head of the platoon **310**.

[0146] For example, in case that the first group includes a third plurality of vehicles exceeding a reference number, the electronic device **100** may identify a first specific vehicle having the largest distance to the branch point among the third plurality of vehicles. The electronic device **100** may control the vehicles **10** such that the first specific vehicle maintains the platooning, and control the vehicles **10** such that remaining vehicles form a sub-platoon in the other lane and subsequently merge at the head of the platoon **310**. For example, in case that the second group includes a fourth plurality of vehicles exceeding the reference number, the electronic device **100** may identify a

second specific vehicle having the largest distance to the branch point among the fourth plurality of vehicles. The electronic device **100** may control vehicles such that the second specific vehicle maintains the platooning in the lane, and may control the vehicles **10** such that remaining vehicles form a sub-platoon in the other lane and subsequently merge at the tail of the platoon.

[0147] FIG. **10** is a flowchart of an electronic device according to an embodiment.

[0148] Each operation of FIG. **10** may be performed sequentially, but is not necessarily performed sequentially. For example, an order of each operation may be changed, and at least two operations may be performed in parallel.

[0149] Referring to FIG. **10**, in operation **1010**, an electronic device **100** may identify a vehicle having the closest distance to a branch point among vehicles **10**.

[0150] In operation **1020**, the electronic device **100** may confirm whether there is a section including multiple lanes in a route to the branch point of the closest vehicle.

[0151] As a result of a determination of the operation **1020**, in case that there is the section including the multiple lanes (operation **1030**: YES), operation **1060** may be performed, and otherwise (operation **1030**: NO), operation **1040** may be performed.

[0152] In the operation **1040**, the electronic device **100** may determine a safety speed for the closest vehicle to leave a platoon **310** at the branch point.

[0153] In operation **1050**, the electronic device **100** may control the vehicles **10** such that the vehicles in the platoon **310** travel at the safe speed until the closest vehicle leaves.

[0154] In operation **1060**, the electronic device **100** may rearrange an order of the vehicles at the section including the multiple lanes.

[0155] FIG. **11** illustrates an example of a block diagram illustrating an autonomous driving system of a vehicle according to an embodiment.

[0156] The autonomous driving system **1100** of the vehicle according to FIG. **11** may be a deep learning network including sensors **1103**, an image pre-processor **1105**, a deep learning network **1107**, an artificial intelligence (AI) processor **1109**, a vehicle control module **1111**, a network interface **1113**, and a communication unit **1115**. In various embodiments, each element may be connected through various interfaces. For example, sensor data sensed and outputted by the sensors **1103** may be fed to the image pre-processor **1105**. The sensor data processed by the image pre-processor **1105** may be fed to the deep learning network **1107** running on the AI processor **1109**. An output of the deep learning network **1107** running by the AI processor **1109** may be fed to the vehicle control module **1111**. Intermediate results of the deep learning network **1107** running on the AI processor **1109** may be fed to the AI processor **1109**. In various embodiments, the network interface **1113** delivers autonomous driving route information and/or autonomous driving control commands for autonomous driving of the vehicle to internal block configurations, by performing communication with an electronic device (e.g., the electronic device **100** and/or the other electronic devices **200** of FIG. **2**) in the vehicle. In an embodiment, the network interface **1113** may be used to transmit the sensor data obtained through the sensor(s) **1103** to an external server. In some embodiments, the autonomous driving control system **1100** may include additional or fewer components as appropriate. For example, in some embodiments, the image pre-processor **1105** may be an optional component. For another example, a post-processing component (not illustrated) may be included in the autonomous driving control system **1100** to perform post-processing on the output of the deep learning network **1107** before the output is provided to the vehicle control module **1111**.

[0157] In some embodiments, the sensors **1103** may include one or more sensors. In various embodiments, the sensors **1103** may be attached to different locations of the vehicle. The sensors **1103** may face one or more different directions. For example, the sensors **1103** may be attached to a front, sides, a rear, and/or a roof of the vehicle to face directions such as forward-facing, rear-facing, and side-facing. In some embodiments, the sensors **1103** may be image sensors such as high dynamic range cameras. In some embodiments, the sensors **1103** include non-visual sensors. In



some embodiments, the sensors **1103** include RADAR, Light Detection And Ranging (LiDAR), and/or ultrasonic sensors in addition to an image sensor. In some embodiments, the sensors **1103** are not mounted on a vehicle having the vehicle control module **1111**. For example, the sensors **1103** may be included as a portion of a deep learning system for capturing the sensor data and may be attached to an environment or a roadway and/or mounted on nearby vehicles.

[0158] In some embodiments, the image pre-processor **1105** may be used to pre-process the sensor data of the sensors **1103**. For example, the image pre-processor **1105** may be used to preprocess the sensor data, to split the sensor data into one or more components, and/or to post-process one or more components. In some embodiments, the image pre-processor **1105** may be a graphics processing unit (GPU), a central processing unit (CPU), an image signal processor, or a specialized image processor. In various embodiments, the image pre-processor **1105** may be a tone-mapper processor for processing high dynamic range data. In some embodiments, the image pre-processor **1105** may be a component of the AI processor **1109**.

[0159] In some embodiments, the deep learning network **1107** may be a deep learning network for implementing control commands for controlling an autonomous vehicle. For example, the deep learning network **1107** may be an artificial neural network such as a convolution neural network (CNN) trained by using the sensor data, and the output of the deep learning network **1107** is provided to the vehicle control module **1111**.

[0160] In some embodiments, the artificial intelligence (AI) processor **1109** may be a hardware processor for running the deep learning network **1107**. In some embodiments, the AI processor **1109** is a specialized AI processor for performing inference on the sensor data through the convolution neural network (CNN). In some embodiments, the AI processor **1109** may be optimized for a bit depth of the sensor data. In some embodiments, the AI processor **1109** may be optimized for deep learning computations, such as computations of a neural network including a convolution, a dot product, a vector and/or matrix computations. In some embodiments, the AI processor **1109** may be implemented through a plurality of graphics processing units (GPUs) capable of effectively performing parallel processing.

[0161] In various embodiments, the AI processor **1109** may be coupled, through an input/output interface, to memory configured to perform a deep learning analysis on the sensor data received from the sensor(s) **1103** while the AI processor **1109** is running and to provide an AI processor having commands that cause to determine a machine learning result used to operate the vehicle at least partially autonomously. In some embodiments, the vehicle control module **1111** may be used to process commands for vehicle control outputted from the artificial intelligence (AI) processor **1109** and translate the output of the AI processor **1109** into commands for controlling a module of each vehicle to control various modules of the vehicle. In some embodiments, the vehicle control module **1111** is used to control a vehicle for autonomous driving. In some embodiments, the vehicle control module **1111** may adjust steering and/or speed of the vehicle. For example, the vehicle control module **1111** may be used to control traveling of the vehicle such as deceleration, acceleration, steering, lane change, lane keeping, and the like. In some embodiments, the vehicle control module **1111** may generate control signals for controlling vehicle lighting, such as brake lights, turns signals, headlights, and the like. In some embodiments, the vehicle control module **1111** may be used to control vehicle audio-related systems such as a vehicle's sound system, vehicle's audio warnings, a vehicle's microphone system, a vehicle's horn system, and the like.

[0162] In some embodiments, the vehicle control module **1111** may be used to control notification systems, including warning systems to notify passengers and/or a driver of driving events, such as approach of an intended destination or a potential collision. In some embodiments, the vehicle control module **1111** may be used to adjust sensors, such as the sensors **1103** of the vehicle. For example, the vehicle control module **1111** may modify the orientation of the sensors **1103**, change output resolution and/or a format type of the sensors **1103**, increase or decrease a capture rate, adjust a dynamic range, and adjust a focus of the camera. In addition, the vehicle control module

**1111** may turn on/off the operation of sensors individually or collectively.

[0163] In some embodiments, the vehicle control module **1111** may be used to change parameters of the image pre-processor **1105** in a method such as modifying a frequency range of filters, adjusting features and/or edge detection parameters for object detection, or adjusting channels and a bit depth, and the like. In various embodiments, the vehicle control module **1111** may be used to control autonomous driving of the vehicle and/or a driver assistance function of the vehicle.

[0164] In some embodiments, the network interface **1113** may be responsible for an internal interface between block configurations of the autonomous driving control system **1100** and the communication unit **1115**. Specifically, the network interface **1113** may be a communication interface for receiving and/or transmitting data including voice data. According to various embodiments, the network interface **1113** may be connected to external servers to connect voice calls, receive and/or transmit text messages, transmit sensor data, update software of the vehicle with the autonomous driving system, or update software of the autonomous driving system of the vehicle, through the communication unit **1115**.

[0165] In various embodiments, the communication unit **1115** may include various wireless interfaces of cellular or WiFi methods. For example, the network interface **1113** may be used to receive an update on operating parameters and/or commands for the sensors **1103**, the image pre-processor **1105**, the deep learning network **1107**, the AI processor **1109**, and the vehicle control module **1111** from an external server connected through the communication unit **1115**. For example, a machine learning model of the deep learning network **1107** may be updated by using the communication unit **1115**. According to another example, the communication unit **1115** may be used to update operating parameters of the image pre-processor **1105**, such as image processing parameters, and/or firmware of the sensors **1103**.

[0166] In another embodiment, the communication unit **1115** may be used to activate communications for an emergency contact and emergency services in an accident or near-accident event. For example, in a crash event, the communication unit **1115** may be used to call emergency services for assistance and may be used to externally notify emergency services of crash details and a location of the vehicle. In various embodiments, the communication unit **1115** may update or obtain an expected arrival time and/or a destination location.

[0167] According to an embodiment, the autonomous driving system **1100** illustrated in FIG. **11** may be configured with an electronic device **100** of the vehicle. According to an embodiment, when an autonomous driving release event occurs from a user during autonomous driving of the vehicle, the AI processor **1109** of the autonomous driving system **1100** may control the software of the vehicle autonomous driving to learn by controlling information related to the autonomous driving release event to be inputted as training set data of the deep learning network.

[0168] FIGS. **12** and **13** illustrate an example of a block diagram indicating an autonomous driving moving object according to an embodiment. FIG. **14** illustrates an example of a gateway related to a user device according to various embodiments.

[0169] Referring to FIG. **12**, an autonomous moving object **1200** according to the present embodiment may include a control device **1300**, sensing modules **1204a**, **1204b**, **1204c**, and **1204d**, an engine **1206**, and a user interface **1208**.

[0170] The autonomous driving moving object **1200** may have an autonomous driving mode or a manual mode. As an example, according to a user input received through the user interface **1208**, it may be switched from the manual mode to the autonomous driving mode or may be switched from the autonomous driving mode to the manual mode.

[0171] In case that the moving object **1200** operates in the autonomous driving mode, the autonomous driving moving object **1200** may operate under control of the control device **1300**.

[0172] In the present embodiment, the control device **1300** may include a controller **1320**, including memory **1322** and a processor **1324**, a sensor **1310**, a communication device **1330**, and an object detection device **1340**.

[0173] Herein, the object detection device **1340** may perform all or a portion of a function of a distance measurement device.

[0174] That is, in the present embodiment, the object detection device **1340** is a device for detecting an object located outside the moving object **1200**, and the object detection device **1340** may detect the object located outside the moving object **1200** and generate object information according to the detection result.

[0175] The object information may include information on existence or nonexistence of the object, location information of the object, distance information between the moving object and the object, and relative speed information between the moving object and the object.

[0176] The object may include various objects located outside the moving object **1200**, such as a lane, another vehicle, a pedestrian, a traffic signal, light, a road, a structure, a speed bump, a landform, an animal, and the like. Herein, the traffic signal may be a concept including a traffic signal, a traffic sign, a pattern or text drawn on a road surface. In addition, the light may be light generated from a lamp equipped in another vehicle, light generated from a streetlamp, or sunlight.

[0177] In addition, the structure may be an object located around a road and fixed to the ground. For example, the structure may include a streetlamp, a street tree, a building, a power pole, a traffic light, and a bridge. The landform may include a mountain, a hill, and the like.

[0178] Such the object detection device **1340** may include a camera module. The controller **1320** may extract object information from an external image photographed by the camera module and enable the controller **1320** to process information thereon.

[0179] In addition, the object detection device **1340** may further include imaging devices for recognizing an external environment. RADAR, a GPS device, Odometry, and another computer vision device, an ultrasonic sensor, and an infrared sensor may be used, in addition to LIDAR, and these devices may be selected or operated simultaneously as needed to enable more precise detection.

[0180] Meanwhile, the distance measurement device according to an embodiment of the present invention may calculate a distance between the autonomous driving moving object **1200** and the object, and may control an operation of the moving object based on the distance calculated in connection with the control device **1300** of the autonomous driving moving object **1200**.

[0181] As an example, in case that there is a probability of a collision according to the distance between the autonomous driving moving object **1200** and the object, the autonomous driving moving object **1200** may control a brake to lower a speed or stop. As another example, in case that the object is a moving object, the autonomous driving moving object **1200** may control a traveling speed of the autonomous driving moving object **1200** to maintain a predetermined distance or more from the object.

[0182] This distance measurement device according to an embodiment of the present invention may be configured as a module in the control device **1300** of the autonomous driving moving object **1200**. That is, the memory **1322** and the processor **1324** of the control device **1300** may be configured to implement a collision prevention method according to the present invention in software.

[0183] In addition, the sensor **1310** may obtain various sensing information by connecting an internal/external environment of the moving object with the sensing modules **1204a**, **1204b**, **1204c**, and **1204d**. Herein, the sensor **1310** may include a posture sensor, a yaw sensor, a roll sensor, a pitch sensor, a collision sensor, a wheel sensor, a speed sensor, a tilt sensor, a weight detection sensor, a heading sensor, a gyro sensor, a position module, a moving object forward/rearward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor by handle rotation, a moving object internal temperature sensor, a moving object internal humidity sensor, an ultrasonic sensor, an illumination sensor, an accelerator pedal position sensor, a brake pedal position sensor, and the like.

[0184] Accordingly, the sensor **1310** may obtain sensing signals for moving object posture

information, moving object collision information, moving object direction information, moving object location information (GPS information), moving object angle information, moving object speed information, moving object acceleration information, moving object tilt information, moving object forward/rearward information, battery information, fuel information, tire information, moving object lamp information, and moving object internal temperature information, moving object internal humidity information, a steering wheel rotation angle, moving object external illumination, a pressure applied to an accelerator pedal, a pressure applied to a brake pedal, and the like.

[0185] In addition, the sensor **1310** may further include an accelerator pedal sensor, a pressure sensor, an engine speed sensor, an air flow sensor (AFS), an intake air temperature sensor (ATS), a water temperature sensor (WTS), a throttle position sensor (TPS), a TDC sensor, a crank angle sensor (CAS), and the like.

[0186] As such, the sensor **1310** may generate moving object state information based on sensing data.

[0187] The wireless communication device **1330** is configured to implement wireless communication between the autonomous driving moving object **1200**. For example, The wireless communication device **1330** enables the autonomous driving moving object **1200** to communicate with a mobile phone of a user, or the other wireless communication device **1330**, another moving object, a central device (a traffic control device), a server, and the like. The wireless communication device **1330** may transmit and receive a wireless signal according to an access wireless protocol. A wireless communication protocol may be Wi-Fi, Bluetooth, Long-Term Evolution (LTE), Code Division Multiple Access (CDMA), Wideband Code Division Multiple Access (WCDMA), Global Systems for Mobile Communications (GSM), but the communication protocol is not limited thereto.

[0188] In addition, in the present embodiment, it is also possible for the autonomous driving moving object **1200** to implement communication between moving objects through the wireless communication device **1330**. That is, the wireless communication device **1330** may perform communication with another moving object and other moving objects on the road through vehicle-to-vehicle (V2V) communication. The autonomous driving moving object **1200** may transmit and receive information such as driving warning and traffic information through the vehicle-to-vehicle (V2V) communication, and it is also possible to request information from, or receive a request from the other moving object. For example, the wireless communication device **1330** may perform the V2V communication as a dedicated short-range communication (DSRC) device or a Cellular-V2V (C-V2V) device. In addition, besides the vehicle-to-vehicle (V2V) communication, communication (e.g., Vehicle to Everything communication (V2X)) between a vehicle and another object (e.g., an electronic device carried by a pedestrian, and the like) may also be implemented through the wireless communication device **1330**.

[0189] In addition, the wireless communication device **1330** may obtain information generated from various mobilities, including infrastructure (a traffic light, a CCTV, a RSU, a eNode B, and the like) located on the road or other autonomous driving/non-autonomous driving vehicles, and the like, through a non-terrestrial network other than a terrestrial network, as information for autonomous driving performance of the autonomous driving moving object **1200**.

[0190] For example, the wireless communication device **1330** may perform wireless communication through a Low Earth Orbit (LEO) satellite system, a Medium Earth Orbit (MEO) satellite system, a Geostationary Orbit (GEO) satellite system, a High Altitude Platform (HAP) system, and the like, that configure a non-terrestrial network and an antenna dedicated to the non-terrestrial network mounted on the autonomous driving moving object **1200**.

[0191] For example, the wireless communication device **1330** may perform wireless communication with various platforms configuring the NTN according to a 5TH Generation New Radio Non-Terrestrial Network (5G NR NTN) standard, which is currently discussed in 3GPP, and

the like, but is not limited thereto.

[0192] In the present embodiment, the controller **1320** may select a platform that may properly perform NTN communication in consideration of various information such as a location of the autonomous driving moving object **1200**, current time, and available power, and control the wireless communication device **1330** to perform wireless communication with the selected platform.

[0193] In the present embodiment, the controller **1320**, which is a unit that controls an overall operation of each unit in the moving object **1200**, may be configured by a manufacturer of the moving object when manufacturing or may be additionally configured to perform a function of autonomous driving after manufacturing. In addition, a configuration for performing a continuous additional function may be included through an upgrade of the controller **1320** configured when manufacturing. This controller **1320** may also be named an Electronic Control Unit (ECU).

[0194] The controller **1320** may collect various data from the connected sensor **1310**, the object detection device **1340**, the communication device **1330**, and may transmit a control signal to the sensor **1310**, the engine **1206**, the user interface **1208**, the communication device **1330**, and the object detection device **1340** included in other components in the moving object based on the collected data. In addition, although not illustrated, the control signal may also be transmitted to an acceleration device, a braking system, a steering device, or a navigation device related to traveling of the moving object.

[0195] In the present embodiment, the controller **1320** may control the engine **1206**, for example, may detect a speed limit of a road on which the autonomous driving moving object **1200** is traveling, and may control the engine **1206** so that a traveling speed does not exceed the speed limit or may control the engine **1206** to accelerate the traveling speed of the autonomous driving moving object **1200** in a range that does not exceed the speed limit.

[0196] In addition, when the autonomous driving moving object **1200** approaches a lane or leaves the lane while the autonomous driving moving object **1200** is traveling, the controller **1320** may determine whether such lane approaching and leaving are due to a normal traveling situation or another traveling situation, and may control the engine **1206** to control the traveling of the moving object according to the determination result. Specifically, the autonomous driving moving object **1200** may detect lanes formed on both sides of the lane in which the moving object is traveling. In this case, the controller **1320** may determine whether the autonomous driving moving object **1200** approaches the lane or leaves the lane, and if it is determined that the autonomous driving moving object **1200** approaches the lane or leaves the lane, the controller **1320** may determine whether this traveling is according to an accurate traveling situation or another traveling situation. Herein, as an example of the normal traveling situation, it may be a situation in which a lane change of the moving object is required. In addition, as an example of the other driving situations, it may be a situation in which a lane change of the moving object is not required. When it is determined that the autonomous driving moving object **1200** is approaching the lane or leaving the lane in a situation in which the moving object does not need to change lane, the controller **1320** may control the traveling of the autonomous driving moving object **1200** so that the autonomous driving moving object **1200** does not leave the lane and normally travels in a corresponding vehicle.

[0197] In case that another moving object or an obstacle exists in a front of the moving object, it may control the engine **1206** or the braking system to decelerate the driving moving object, and may control a trajectory, a traveling route, and a steering angle in addition to speed. Alternatively, the controller **1320** may control the traveling of the moving object by generating a necessary control signal according to recognition information of another external environment, such as a traveling lane or a driving signal of the moving object.

[0198] In addition to generating its own control signal, the controller **1320** may also control the traveling of the moving object by performing communication with a nearby moving object or a central server and transmitting a command to control peripheral devices through the received

information.

[0199] In addition, since accurate recognition of the moving object or lane according to the present embodiment may be difficult in case that a location of the camera module **1350** changes or an angle of view changes, the controller **1320** may generate a control signal for controlling to perform calibration of the camera module **1350** to prevent this. Therefore, in the present embodiment, by generating the calibration control signal to the camera module **1350**, the controller **1320** may continuously maintain a normal mounting location, a direction, an angle of view, and the like of the camera module **1350** even when a mounting location of the camera module **1350** is changed due to vibration or impact generated by a movement of the autonomous driving moving object **1200**. In case that an initial mounting location, a direction, and an angle of view information of the camera module **1350** that are pre-stored, and an initial mounting location, a direction, an angle of view information, and the like of the camera module **1350** measured while the autonomous driving moving object **800** is traveling are changed by a threshold value or more, the controller **1320** may generate the control signal to perform the calibration of the camera module **1350**.

[0200] In the present embodiment, the controller **1320** may include the memory **1322** and the processor **1324**. The processor **1324** may execute software stored in the memory **1322** according to the control signal of the controller **1320**. Specifically, the controller **1320** may store data and commands for performing the lane detection method according to the present invention in the memory **1322**, and the commands may be executed by the processor **1324** to implement one or more methods disclosed herein.

[0201] In this case, the memory **1322** may be stored in a recording medium executable by the non-volatile processor **1324**. The memory **1322** may store software and data through an appropriate internal/external device. The memory **1322** may be configured with random access memory (RAM), read only memory (ROM), a hard disk, and a memory **1322** device connected with a dongle.

[0202] The memory **1322** may at least store an Operating system (OS), a user application, and executable commands. The memory **1322** may also store application data and array data structures.

[0203] The processor **1324**, which is a microprocessor or an appropriate electronic processor, may be a controller, a microcontroller, or a state machine.

[0204] The processor **1324** may be implemented as a combination of computing devices, and the computing device may be configured with a digital signal processor, a microprocessor, or an appropriate combination thereof.

[0205] Meanwhile, the autonomous driving moving object **1200** may further include the user interface **1208** for a user input with respect to the above-described control device **1300**. The user interface **1208** may enable a user to input information with appropriate interaction. For example, it may be implemented as a touch screen, a keypad, or an operation button, and the like. The user interface **1208** may transmit an input or a command to the controller **1320**, and the controller **1320** may perform a control operation of the moving object in response to the input or the command.

[0206] In addition, the user interface **1208**, which is a device outside the autonomous driving moving object **1200**, may perform communication with the autonomous driving moving object **1200** through the wireless communication device **1330**. For example, the user interface **808** may be linkable with a mobile phone, a tablet, or another computer device.

[0207] Furthermore, in the present embodiment, the autonomous driving moving object **1200** has been described as including the engine **1206**, but it may also include another type of a propulsion system. For example, the moving object may be operated with electrical energy, and may be operated through hydrogen energy or a hybrid system combining them. Therefore, the controller **1320** may include a propulsion mechanism according to the propulsion system of the autonomous driving moving object **1200** and may provide a control signal according to this to components of each propulsion mechanism.

[0208] Hereinafter, a detailed configuration of the control device **1300** according to the present

invention according to the present embodiment will be described in more detail with reference to FIG. 14.

[0209] A control device **1300** includes a processor **1324**. The processor **1324** may be a general-purpose single or multi-chip microprocessor, a dedicated microprocessor, a microcontroller, a programmable gate array, and the like. The processor may be referred to as a central processing unit (CPU). In addition, in the present embodiment, it is possible that the processor **1324** is used as a combination of a plurality of processors.

[0210] The control device **1300** also includes memory **1322**. The memory **1322** may be any electronic component capable of storing electronic information. The memory **1322** may also include a combination of the memories **1322** in addition to single memory.

[0211] Data and commands **1322a** for performing a distance measuring method of a distance measuring device according to the present invention may be stored in the memory **1322**. When the processor **1324** executes the commands **1322a**, all or a portion of the commands **1322a** and the data **1322b** required for performing a command may be loaded **1324a** and **1324b** onto the processor **1324**.

[0212] The control device **1300** may include a transmitter **1330a**, a receiver **1330b**, or a transceiver **1330c** for permitting transmission and reception of signals. One or more antennas **1332a** and **1332b** may be electrically connected to the transmitter **1330a**, the receiver **1330b**, or each transceiver **1330c**, and may further include antennas.

[0213] The control device **1300** may include a digital signal processor (DSP) **1370**. Through the DSP **1370**, the digital signal may be quickly processed by a moving object.

[0214] The control device **1300** may include a communication interface **1380**. The communication interface **1380** may include one or more ports and/or communication modules for connecting other devices to the control device **1300**. The communication interface **1380** may enable a user and the control device **1300** to interact with each other.

[0215] Various configurations of the control device **1300** may be connected together by one or more buses **1390**, and the buses **1390** may include a power bus, a control signal bus, a state signal bus, a data bus, and the like. Under a control of the processor **1324**, configurations may transmit mutual information through the bus **1390** and perform a desired function.

[0216] Meanwhile, in various embodiments, the control device **1300** may be related to a gateway for communication with a security cloud. For example, referring to FIG. 14, the control device **1300** may be related to a gateway **1405** for providing information obtained from at least one of components **1001** to **1004** of a vehicle **1400** to a security cloud **1406**. For example, the gateway **1405** may be included in the control device **1300**. For another example, the gateway **1405** may be configured as a separate device in the vehicle **1400** that is distinguished from the control device **1300**. The gateway **1405** connects a network in the vehicle **1400** secured by a software management cloud **1409**, the security cloud **1406**, and in-car security software **1410**, having different networks, to enable communication.

[0217] For example, a component **1401** may be a sensor. For example, the sensor may be used to obtain information on at least one of a state of the vehicle **1400** or a state around the vehicle **1400**. For example, the component **1401** may include a sensor **1310**.

[0218] For example, a component **1402** may be electronic control units (ECUs). For example, the ECUs may be used for engine control, transmission control, airbag control, and tire pressure management.

[0219] For example, a component **1403** may be an instrument cluster. For example, the instrument cluster may mean a panel located in a front of a driver's seat among dashboards. For example, the instrument cluster may be configured to display information necessary for driving to a driver (or a passenger). For example, the instrument cluster may be used to display at least one of visual elements for indicating a revolutions per minute (or rotates per minute) (RPM) of the engine, visual elements for indicating a speed of the vehicle **1400**, visual elements for indicating an amount of

remaining fuel, visual elements for indicating a state of a gear, or visual elements for indicating information obtained through the component **1401**.

[0220] For example, a component **1404** may be a telematics device. For example, the telematics device may mean a device that provides various mobile communication services, such as location information and safe driving in the vehicle **1400** by coupling wireless communication technology and global positioning system (GPS) technology. For example, the telematics device may be used to connect the vehicle **1400** with a driver, a cloud (e.g., the security cloud **1406**), and/or a surrounding environment. For example, the telematics device may be configured to support high bandwidth and low latency for 5G NR-standard technology (e.g., V2X technology of the 5G NR, Non-Terrestrial Network (NTN) technology of the 5G NR). For example, the telematics device may be configured to support autonomous driving of the vehicle **1400**.

[0221] For example, the gateway **1405** may be used to connect a network within the vehicle **1400**, and the software management cloud **1409** and the secure cloud **1406**, which are a network outside the vehicle. For example, the software management cloud **1409** may be used to update or manage at least one software necessary for traveling and managing the vehicle **1400**. For example, the software management cloud **1409** may be linked to the in-car security software **1410** installed in the vehicle. For example, the in-car security software **1410** may be used to provide a security function in the vehicle **1400**. For example, the in-car security software **1410** may encrypt data transmitted and received through an in-car network using an encryption key obtained from an external authorized server for encryption of the in-car network. In various embodiments, the encryption key used by the in-car security software **1410** may be generated corresponding to vehicle identification information (a vehicle license plate, a vehicle identification number (VIN)) or information (e.g., user identification information) uniquely assigned to each user.

[0222] In various embodiments, the gateway **1405** may transmit the data encrypted by the in-car security software **1410** based on the encryption key to the software management cloud **1409** and/or the security cloud **1406**. The software management cloud **1409** and/or the security cloud **1406** may identify the data received from which vehicle or which user by decrypting the data encrypted by the encryption key of the in-car security software **1410**. For example, since the decryption key is a unique key corresponding to the encryption key, the software management cloud **1409** and/or the security cloud **1406** may identify a transmission entity (e.g., the vehicle or the user) of the data based on the data decrypted through the decryption key.

[0223] For example, the gateway **1405** may be configured to support in-car security software **1410** and may be related to the control device **1300**. For example, the gateway **1405** may be related to the control device **1300** to support a connection between a client device **1407** and the control device **1300** connected to the security cloud **1406**. For another example, the gateway **1405** may be related to the control device **1300** to support a connection between a third-party cloud **1408** connected to the security cloud **1406** and the control device **1300**. However, it is not limited thereto.

[0224] In various embodiments, the gateway **1405** may be used to connect the vehicle **1400** with the software management cloud **1409** to manage operating software of the vehicle **1400**. For example, the software management cloud **1409** may monitor whether updating the operating software of the vehicle **1400** is required, and based on monitoring that the updating the operating software of the vehicle **1400** is required, provide data for the updating the operating software of the vehicle **1400** through the gateway **1405**. For another example, the software management cloud **1409** may receive a user request for updating the operating software of the vehicle **1400** from the vehicle **1400** through the gateway **1405**, and provide data for updating the operating software of the vehicle **1400** based on the reception. However, it is not limited thereto.

[0225] FIG. **15** is a diagram for explaining an operation of an electronic device for training a neural network based on a set of learning data, according to an embodiment.

[0226] An operation described with reference to FIG. **15** may be performed by the above-described



electronic device (e.g., the electronic device **100** of FIG. 2).

[0227] Referring to FIG. 15, in operation **1502**, the electronic device may obtain the set of the learning data according to an embodiment. The electronic device may obtain the set of the learning data for supervised learning. The learning data may include a pair of input data and ground truth data corresponding to the input data. The ground truth data may indicate output data to be obtained from the neural network that has received the input data, which is the pair of the ground truth data. The ground truth data may be obtained by the electronic device described above.

[0228] For example, in case of training the neural network for image recognition, the learning data may include information regarding an image and one or more subjects included within the image. The information may include a category (or a class) of a subject identifiable through the image. The information may include a location, a width, a height, and/or a size of a visual object corresponding to the subject within the image. The set of the learning data identified through the operation **1502** may include pairs of a plurality of learning data. In the example of training the neural network for the image recognition, the set of the learning data identified by the electronic device may include a plurality of images and ground truth data corresponding to each of the plurality of images.

[0229] Referring to FIG. 15, in operation **1504**, the electronic device according to an embodiment may perform training on the neural network based on the set of the learning data. In an embodiment in which the neural network is trained based on the supervised learning, the electronic device may input the input data included in the learning data to an input layer of the neural network. An example of the neural network including the input layer will be described with reference to FIG. **16**. From an output layer of the neural network receiving the input data through the input layer, the electronic device may obtain output data of the neural network corresponding to the input data.

[0230] In an embodiment, the training of the operation **1504** may be performed based on a difference between the output data and the ground truth data included in the learning data and corresponding to the input data. For example, the electronic device may adjust one or more parameters related to the neural network (e.g., a weight to be described later with reference to FIG. **16**) to reduce the difference based on a gradient descent algorithm. An operation of the electronic device adjusting the one or more parameters may be referred to as tuning for the neural network. The electronic device may perform the tuning of the neural network based on the output data using a function defined to evaluate performance of the neural network, such as a cost function. The difference between the output data and the ground truth data may be included as an example of the cost function.

[0231] Referring to FIG. 15, in operation **1506**, according to an embodiment, the electronic device may identify whether valid output data is outputted from the neural network trained by the operation **1504**. The output data being valid may mean that the difference (or the cost function) between the output data and the ground truth data satisfies a condition set for use of the neural network. For example, in case that an average value and/or the maximum value of the difference between the output data and the ground truth data is less than or equal to a designated threshold value, the electronic device may determine that the valid output data is outputted from the neural network.

[0232] In case that the valid output data is not outputted from the neural network (**1506-NO**), the electronic device may repeatedly perform training of the neural network based on the operation **1504**. An embodiment is not limited thereto, and the electronic device may repeatedly perform the operations **1502** and **1504**.

[0233] In a state in which the valid output data is obtained from the neural network (**1506-YES**), based on operation **1508**, the electronic device according to an embodiment may use the trained neural network. For example, the electronic device may input other input data to the neural network that is distinct from the input data inputted to the neural network as the learning data. The electronic device may use output data obtained from the neural network receiving the other input

data as a result of performing inference on the other input data based on the neural network.

[0234] FIG. **16** is a block diagram of an electronic device according to an embodiment.

[0235] An electronic device **100** of FIG. **16** may include the above-described electronic device.

[0236] For example, an operation described with reference to FIG. **16** may be performed by the electronic device **100** of FIG. **16** and/or a processor **1610** of FIG. **16**.

[0237] Referring to FIG. **16**, the processor **1610** of the electronic device **100** may perform computations related to a neural network **1630** stored in memory **1620**. The processor **1610** may include at least one of a central processing unit (CPU), a graphic processing unit (GPU), and a neural processing unit (NPU). The NPU may be implemented as a chip separated from the CPU, or integrated into a chip such as the CPU in a form of a system on a chip (SoC). The NPU integrated into the CPU may be referred to as a neural core and/or an artificial intelligence (AI) accelerator.

[0238] Referring to FIG. **16**, the processor **1610** may identify the neural network **1630** stored in the memory **1620**. The neural network **1630** may include a combination of an input layer **1632**, one or more hidden layers **1634** (or intermediate layers), and an output layer **1636**. The above-described layers (e.g., the input layer **1632**, the one or more hidden layers **1634**, and the output layer **1636**) may include a plurality of nodes. The number of hidden layers **1634** may vary according to an embodiment, and the neural network **1630** including the plurality of hidden layers **1634** may be referred to as a deep neural network. An operation of training the deep neural network may be referred to as deep learning.

[0239] In an embodiment, in case that the neural network **1630** has a structure of a feed forward neural network, a first node included in a specific layer may be connected to all of second nodes included in another layer before the specific layer. In the memory **1620**, parameters stored for the neural network **1630** may include weights assigned to connections between the second nodes and the first node. In the neural network **1630** having the structure of the feed forward neural network, a value of the first node may correspond to a weighted sum of values assigned to the second nodes, based on the weights assigned to the connections connecting the second nodes and the first node.

[0240] In an embodiment, in case that the neural network **1630** has a structure of a convolutional neural network, the first node included in the specific layer may correspond to a weighted sum of a portion of the second nodes included in the other layer before the specific layer. The portion of the second nodes corresponding to the first node may be identified by a filter corresponding to the specific layer. In the memory **1620**, the parameters stored for the neural network **1630** may include weights indicating the filter. The filter may include, among the second nodes, one or more nodes to be used to calculate a weighted sum of the first node, and weights corresponding to each of the one or more nodes.

[0241] According to an embodiment, the processor **1610** of the electronic device **100** may perform training on the neural network **1630** using a learning data set **1640** stored in the memory **1620**.

Based on the learning data set **1640**, the processor **1610** may adjust one or more parameters stored in the memory **1620** for the neural network **1630** by performing the operation described with reference to FIG. **15**.

[0242] According to an embodiment, the processor **1610** of the electronic device **100** may perform object detection, object recognition, and/or object classification using the neural network **1630** trained based on the learning data set **1640**. The processor **1610** may input an image (or a video) obtained through a camera **1650** into the input layer **1632** of the neural network **1630**. Based on the input layer **1632** to which the image is inputted, the processor **1610** may obtain a set (e.g., the output data) of values of the nodes of the output layer **1636** by sequentially obtaining values of the nodes of the layers included in the neural network **1630**. The output data may be used as a result of inferring information included in the image using the neural network **1630**. An embodiment is not limited thereto, and the processor **1610** may input an image (or a video) obtained from an external electronic device connected to the electronic device **100** through communication circuitry **1660** to the neural network **1630**.

[0243] In an embodiment, the neural network **1630** trained to process an image may be used to identify a region corresponding to a subject within the image (object detection), and/or to identify a class of the subject represented within the image (object recognition and/or object classification). For example, the electronic device **100** may segment the region corresponding to the subject within the image based on a quadrangle shape such as a bounding box, using the neural network **1630**. For example, the electronic device **100** may identify at least one class matching the subject among a plurality of designated classes using the neural network **1630**.

[0244] FIG. **17** is a flowchart of an electronic device according to an embodiment. Each operation of FIG. **17** may be sequentially performed, but is not necessarily performed sequentially. For example, an order of each operation may be changed, and at least two operations may be performed in parallel.

[0245] Hereinafter, operations in which an electronic device **100** obtains control authority for other vehicles in a platoon **310** and controls the other vehicles in the platoon **310** according to the obtained control authority will be described with reference to FIG. **17**.

[0246] In operation **1710**, the electronic device **100** may request the control authority to the other vehicles in the platoon **310**. For example, the electronic device **100** may transmit the request of the control authority to the other vehicles (or their electronic devices) in the platoon **310**. The control authority may include, for example, a temporary control authority such as a floor control.

[0247] In operation **1720**, the electronic device **100** may receive a response with respect to the control authority request from the other vehicles in the platoon **310**. For example, the response with respect to the request for the control authority may include a positive Ack or a negative Ack.

[0248] In operation **1730**, the electronic device **100** may transmit a control signal to one or more vehicles that have obtained the control authority among the other vehicles in the platoon **310**. For example, the electronic device **100** may obtain the control authority for one or more vehicles that have transmitted the positive Ack among the other vehicles in the platoon **310**. The electronic device **100** may transmit the control signal to the one or more vehicles having the control authority. Accordingly, the operations in which the electronic device **100** controls vehicles **10** in the platoon **310** described above may be performed.

[0249] In an embodiment, an electronic device for platooning of vehicles may comprise memory storing instructions and a processor, and the instructions, when executed by the processor, may cause the electronic device to identify branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon, identify distances from current positions of the vehicles to the branch points, determine an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

[0250] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to, based on identifying other vehicles from among the vehicles in which the distances to the branch points are less than a reference value, change the determined order such that the other vehicles are rearranged in the ascending order of the identified distances from the tail.

[0251] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to, based on an insertion sort algorithm for the identified distances, identify at least one vehicle that has to leave the platoon to be rearranged according to the determined order, determine the at least one vehicle as a first group, where a distance to the branch point exceeds the reference value, and/or as a second group, where a distance to the branch point does not exceed the reference value, and control, when the first group includes a first plurality of vehicles, the vehicles such that the first plurality of vehicles form a first sub-platoon in another lane different from a lane in which the platoon is traveling and subsequently merge at a head of the platoon.

[0252] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to control, when the first group includes only one vehicle, the vehicles such that the one vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

[0253] In an embodiment, the instructions, when executed by the processor, cause the electronic device to control, when the second group includes a second plurality of vehicles, the vehicles such that the second plurality of vehicles form a second sub-platoon in the another lane from the lane in which the platoon is traveling and subsequently merge at a tail of the platoon.

[0254] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to control, when the second group includes only one other vehicle, the vehicles such that the one other vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

[0255] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to determine the at least one vehicle as the first group, where the distance to the branch point exceeds the reference value, and/or as the second group, where the distance to the branch point does not exceed the reference value, and when the first group includes a third plurality of vehicles exceeding a reference number, identify a specific vehicle from among the third plurality of vehicles having the largest distance to the branch point, control the vehicles such that the specific vehicle maintains the platooning in the lane, and control the vehicles such that remaining vehicles form a third sub-platoon in the another lane and subsequently merge at a head of the platoon.

[0256] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to determine the at least one vehicle as the first group, where a distance to the branch point exceeds the reference value, and/or as the second group, where a distance to the branch point does not exceed the reference value, and when the second group includes a fourth plurality of vehicles exceeding a reference number, identify a specific vehicle from among the fourth plurality of vehicles having the largest distance to the branch point, control the vehicles such that the specific vehicle maintains the platooning in the lane, and control the vehicles such that remaining vehicles to form a fourth sub-platoon in the another lane and subsequently merge at a tail of the platoon.

[0257] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to identify the closest vehicle to the branch point from among the vehicles, confirm whether there is a section including multiple lanes in a route to the branch point of the closest vehicle, based on confirming that there is no section including the multiple lanes in the route, determine a safe speed for the closest vehicle to leave the platoon at the branch point of the closest vehicle, and control the vehicles such that the vehicles in the platoon travel at the safe speed until the closest vehicle leaves the platoon.

[0258] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to, based on determining that a lead vehicle and a tail vehicle in the current order in the platoon are identical to a lead vehicle and a tail vehicle in the determined order, control the vehicles to maintain the current order of the vehicles.

[0259] In an embodiment, the instructions, when executed by the processor, may cause the electronic device to obtain information regarding destinations of the vehicles from each of the vehicles, and identify the branch points based on the information.

[0260] In an embodiment, a method performed by an electronic device for platooning of vehicles may comprise identifying branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon, identifying distances from current positions of the vehicles to the branch points, determining an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the

determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, controlling the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

[0261] In an embodiment, the method may comprise, based on identifying other vehicles from among the vehicles in which the distances to the branch points are less than a reference value, changing the determined order such that the other vehicles are rearranged in the ascending order of the identified distances from the tail.

[0262] In an embodiment, the method may comprise, based on an insertion sort algorithm for the identified distances, identifying at least one vehicle that has to leave the platoon to be rearranged according to the determined order, determining the at least one vehicle as a first group, where a distance to the branch point exceeds the reference value, and/or as a second group, where a distance to the branch point does not exceed the reference value, and controlling, when the first group includes a first plurality of vehicles, the vehicles such that the first plurality of vehicles form a first sub-platoon in another lane different from a lane in which the platoon is traveling and subsequently merge at a head of the platoon.

[0263] In an embodiment, the method may comprise controlling, when the first group includes only one vehicle, the vehicles such that the one vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

[0264] In an embodiment, the method may comprise controlling, when the second group includes a second plurality of vehicles, the vehicles such that the second plurality of vehicles form a second sub-platoon in the another lane from the lane in which the platoon is traveling and subsequently merge at a tail of the platoon.

[0265] In an embodiment, the method may comprise controlling, when the second group includes only one other vehicle, the vehicles, such that the one other vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

[0266] In an embodiment, the method may comprise determining the at least one vehicle as the first group, where the distance to the branch point exceeds the reference value, and/or as the second group, where the distance to the branch point does not exceed the reference value, and when the first group includes a third plurality of vehicles exceeding a reference number, identifying a specific vehicle from among the third plurality of vehicles having the largest distance to the branch point, controlling the vehicles, such that the specific vehicle maintains the platooning in the lane, and controlling the vehicles, such that remaining vehicles form a third sub-platoon in the another lane and subsequently merge at a head of the platoon.

[0267] In an embodiment, the method may comprise determining the at least one vehicle as the first group, where a distance to the branch point exceeds the reference value, and/or as the second group, where a distance to the branch point does not exceed the reference value, and when the second group includes a fourth plurality of vehicles exceeding a reference number, identifying a specific vehicle from among the fourth plurality of vehicles having the largest distance to the branch point, controlling the vehicles such that the specific vehicle maintains the platooning in the lane, and controlling the vehicles such that remaining vehicles to form a fourth sub-platoon in the another lane and subsequently merge at a tail of the platoon.

[0268] In an embodiment, a computer-readable medium may store one or more programs. The one or more programs, when executed by an electronic device for platooning of vehicles, may cause the electronic device to identify branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon, identify distances from current positions of the vehicles to the branch points, determine an order of the vehicles in the platoon based on an ascending order of the identified distances, and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the

vehicles in the platoon are rearranged according to the determined order.

[0269] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” or “connected with” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0270] In the above-described specific embodiments of the present disclosure, components included in the disclosure are expressed in the singular or plural according to the presented specific embodiment. However, the singular or plural expression is selected appropriately according to a situation presented for convenience of explanation, and the present disclosure is not limited to the singular or plural component, and even components expressed in the plural may be configured in the singular, or a component expressed in the singular may be configured in the plural.

[0271] According to various embodiments, one or more components or operations of the above-described components may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be executed sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0272] Meanwhile, specific embodiments have been described in the detailed description of the present disclosure, and of course, various modifications are possible without departing from the scope of the present disclosure.

## Claims

1. An electronic device for platooning of vehicles, comprising: memory storing instructions; and a processor, wherein the instructions, when executed by the processor, cause the electronic device to: identify branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon; identify distances from current positions of the vehicles to the branch points; determine an order of the vehicles in the platoon based on an ascending order of the identified distances; and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

2. The electronic device of claim 1, wherein the instructions, when executed by the processor,

cause the electronic device to: based on identifying other vehicles from among the vehicles in which the distances to the branch points are less than a reference value, change the determined order such that the other vehicles are rearranged in the ascending order of the identified distances from the tail.

**3.** The electronic device of claim 2, wherein the instructions, when executed by the processor, cause the electronic device to: based on an insertion sort algorithm for the identified distances, identify at least one vehicle that has to leave the platoon to be rearranged according to the determined order; determine the at least one vehicle as a first group, where a distance to the branch point exceeds the reference value, and/or as a second group, where a distance to the branch point does not exceed the reference value; and control, when the first group includes a first plurality of vehicles, the vehicles such that the first plurality of vehicles form a first sub-platoon in another lane different from a lane in which the platoon is traveling and subsequently merge at a head of the platoon.

**4.** The electronic device of claim 3, wherein the instructions, when executed by the processor, cause the electronic device to: control, when the first group includes only one vehicle, the vehicles such that the one vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

**5.** The electronic device of claim 3, wherein the instructions, when executed by the processor, cause the electronic device to: control, when the second group includes a second plurality of vehicles, the vehicles such that the second plurality of vehicles form a second sub-platoon in the another lane from the lane in which the platoon is traveling and subsequently merge at a tail of the platoon.

**6.** The electronic device of claim 3, wherein the instructions, when executed by the processor, cause the electronic device to: control, when the second group includes only one other vehicle, the vehicles such that the one other vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

**7.** The electronic device of claim 3, wherein the instructions, when executed by the processor, cause the electronic device to: determine the at least one vehicle as the first group, where the distance to the branch point exceeds the reference value, and/or as the second group, where the distance to the branch point does not exceed the reference value; and when the first group includes a third plurality of vehicles exceeding a reference number: identify a specific vehicle from among the third plurality of vehicles having the largest distance to the branch point; control the vehicles such that the specific vehicle maintains the platooning in the lane; and control the vehicles such that remaining vehicles form a third sub-platoon in the another lane and subsequently merge at a head of the platoon.

**8.** The electronic device of claim 3, wherein the instructions, when executed by the processor, cause the electronic device to: determine the at least one vehicle as the first group, where a distance to the branch point exceeds the reference value, and/or as the second group, where a distance to the branch point does not exceed the reference value; and when the second group includes a fourth plurality of vehicles exceeding a reference number: identify a specific vehicle from among the fourth plurality of vehicles having the largest distance to the branch point; control the vehicles such that the specific vehicle maintains the platooning in the lane; and control the vehicles such that remaining vehicles to form a fourth sub-platoon in the another lane and subsequently merge at a tail of the platoon.

**9.** The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: identify the closest vehicle to the branch point from among the vehicles; confirm whether there is a section including multiple lanes in a route to the branch point of the closest vehicle; based on confirming that there is no section including the multiple lanes in the route, determine a safe speed for the closest vehicle to leave the platoon at the branch point of the closest vehicle; and control the vehicles such that the vehicles in the platoon travel at the safe

speed until the closest vehicle leaves the platoon.

**10.** The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: based on determining that a lead vehicle and a tail vehicle in the current order in the platoon are identical to a lead vehicle and a tail vehicle in the determined order, control the vehicles to maintain the current order of the vehicles.

**11.** The electronic device of claim 1, wherein the instructions, when executed by the processor, cause the electronic device to: obtain information regarding destinations of the vehicles from each of the vehicles; and identify the branch points based on the information.

**12.** A method performed by an electronic device for platooning of vehicles, comprising: identifying branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon; identifying distances from current positions of the vehicles to the branch points; determining an order of the vehicles in the platoon based on an ascending order of the identified distances; and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, controlling the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

**13.** The method of claim 12, comprising: based on identifying other vehicles from among the vehicles in which the distances to the branch points are less than a reference value, changing the determined order such that the other vehicles are rearranged in the ascending order of the identified distances from the tail.

**14.** The method of claim 13, comprising: based on an insertion sort algorithm for the identified distances, identifying at least one vehicle that has to leave the platoon to be rearranged according to the determined order; determining the at least one vehicle as a first group, where a distance to the branch point exceeds the reference value, and/or as a second group, where a distance to the branch point does not exceed the reference value; and controlling, when the first group includes a first plurality of vehicles, the vehicles such that the first plurality of vehicles form a first sub-platoon in another lane different from a lane in which the platoon is traveling and subsequently merge at a head of the platoon.

**15.** The method of claim 14, comprising: controlling, when the first group includes only one vehicle, the vehicles such that the one vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

**16.** The method of claim 15, comprising: controlling, when the second group includes a second plurality of vehicles, the vehicles such that the second plurality of vehicles form a second sub-platoon in the another lane from the lane in which the platoon is traveling and subsequently merge at a tail of the platoon.

**17.** The method of claim 14, comprising: controlling, when the second group includes only one other vehicle, the vehicles such that the one other vehicle leaves the lane in which the platoon is traveling and subsequently merges at a head of the platoon.

**18.** The method of claim 14, comprising: determining the at least one vehicle as the first group, where the distance to the branch point exceeds the reference value, and/or as the second group, where the distance to the branch point does not exceed the reference value; and when the first group includes a third plurality of vehicles exceeding a reference number: identifying a specific vehicle from among the third plurality of vehicles having the largest distance to the branch point; controlling the vehicles such that the specific vehicle maintains the platooning in the lane; and controlling the vehicles such that remaining vehicles form a third sub-platoon in the another lane and subsequently merge at a head of the platoon.

**19.** The method of claim 14, comprising: determining the at least one vehicle as the first group, where a distance to the branch point exceeds the reference value, and/or as the second group, where a distance to the branch point does not exceed the reference value; and when the second group



includes a fourth plurality of vehicles exceeding a reference number: identifying a specific vehicle from among the fourth plurality of vehicles having the largest distance to the branch point; controlling the vehicles such that the specific vehicle maintains the platooning in the lane; and controlling the vehicles such that remaining vehicles to form a fourth sub-platoon in the another lane and subsequently merge at a tail of the platoon.

**20.** A computer-readable medium storing one or more programs, wherein the one or more programs, when executed by an electronic device for platooning of vehicles, cause the electronic device to: identify branch points for each of the vehicles on a route of the platooning, wherein the branch points are locations where the vehicles leave the platoon; identify distances from current positions of the vehicles to the branch points; determine an order of the vehicles in the platoon based on an ascending order of the identified distances; and based on determining that a lead vehicle in a current order of the vehicles in the platoon is different from a lead vehicle in the determined order and/or determining that a tail vehicle in the current order of the vehicles in the platoon is different from a tail vehicle in the determined order, control the vehicles such that the vehicles in the platoon are rearranged according to the determined order.

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